

CODEFEST- 4CMAN- 4

Given 4×4 Matrix FP32

$$W = \begin{bmatrix} 0.85 & -1.20 & 0.34 & 2.10 \\ -0.07 & 0.91 & -1.88 & 0.12 \\ 1.55 & 0.03 & -0.44 & -2.31 \\ -0.18 & 1.03 & 0.77 & 0.55 \end{bmatrix}$$

1. Scale Factor:

$$S = \frac{\max |W|}{127}$$

$$|W| = \begin{bmatrix} 0.85 & 1.20 & 0.34 & 2.10 \\ 0.07 & 0.91 & 1.88 & 0.12 \\ 1.55 & 0.03 & 0.44 & 2.31 \\ 0.18 & 1.03 & 0.77 & 0.55 \end{bmatrix}$$

Max element from the given weighted matrix $|W| = 2.31$

$$S = \frac{\max |W|}{127} \Rightarrow \frac{2.31}{127}$$

$$S = 0.0181889763779$$

2. Quantize: $W-q = \text{round}\left(\frac{W}{S}\right)$

Dividing every element of W with S from task 1 and rounding it.

$$W-q = \begin{bmatrix} 0.85/S & -1.20/S & 0.34/S & 2.10/S \\ -0.07/S & 0.91/S & -1.88/S & 0.12/S \\ 1.55/S & 0.03/S & -0.44/S & -2.31/S \\ -0.18/S & 1.03/S & 0.77/S & 0.55/S \end{bmatrix}$$

$$W-q = \begin{bmatrix} 47 & -66 & 19 & 115 \\ -4 & 50 & -103 & 7 \\ 85 & 2 & -24 & -127 \\ -10 & 57 & 42 & 30 \end{bmatrix}$$

3. Dequantize: $W\text{-deg} = W_q \times S$

Now for dequantizing multiplying every element of W_q with S from task 1 and 2.

$$W_q \times S \Rightarrow \begin{bmatrix} 47 \times S & -66 \times S & 19 \times S & 115 \times S \\ -4 \times S & 50 \times S & -103 \times S & 7 \times S \\ 85 \times S & 2 \times S & -24 \times S & -127 \times S \\ -10 \times S & 57 \times S & 42 \times S & 30 \times S \end{bmatrix}$$

$$W\text{-deg} = \begin{bmatrix} 0.8548818897637 & -1.2004724409448 & 0.345590551181 & 2.0917322834645 \\ -0.0727559055118 & 0.9094488188976 & -1.8734645669291 & 0.1273228346456 \\ 1.5460629921259 & 0.0363779527559 & -0.4365354330700 & -2.31 \\ -0.1818897637795 & 1.0367716535433 & 0.7639370078740 & 0.5456692913305 \end{bmatrix}$$

4. Error Analysis: $|W - W\text{-deg}|$

subtracting $W\text{-deg}$ from task 3 with W .

$$\Rightarrow \begin{bmatrix} 0.004881889763 & 0.0004724409448 & 0.005590551181 & 0.0082677165355 \\ 0.002755905118 & 0.0005511811024 & 0.006535433070 & 0.007322834645 \\ 0.0039370078741 & 0.006377952755 & 0.003464566929 & 0 \\ 0.001889763779 & 0.006779527559 & 0.006062992126 & 0.0043307086615 \end{bmatrix}$$

Largest error = 0.0082677165355

Mean Absolute Error (MAE) = Sum of all elements of $|W - W\text{-deg}|$ with 16.

$$\Rightarrow \frac{0.0692204720433}{16}$$

$$\Rightarrow 0.0043262795027$$

5. Bad Scale Experiment: $S_{\text{bad}} = 0.01$

Quantizing, $W_{\text{qbad}} = \text{round}\left(\frac{W}{S_{\text{bad}}}\right)$

$$W_{\text{qbad}} = \begin{bmatrix} 85 & -120 & 34 & 210 \\ -7 & 91 & -188 & 12 \\ 155 & 3 & -44 & -231 \\ -18 & 103 & 77 & 55 \end{bmatrix}$$

After clamping the values to $[-128, 127]$ range.

$$W_{\text{qbad}} = \begin{bmatrix} 85 & -120 & 34 & 127 \\ -7 & 91 & -128 & 12 \\ 127 & 3 & -44 & -128 \\ -18 & 103 & 77 & 55 \end{bmatrix}$$

Dequantizing, ~~W_{q}~~ $W_{\text{degbad}} = W_{\text{qbad}} \times S_{\text{bad}}$

$$W_{\text{degbad}} = \begin{bmatrix} 0.85 & -1.20 & 0.34 & 1.27 \\ -0.07 & 0.91 & -1.28 & 0.12 \\ 1.27 & 0.03 & -0.44 & -1.28 \\ -0.18 & 1.03 & 0.77 & 0.55 \end{bmatrix}$$

Absolute error, $|W - W_{\text{degbad}}|$

$$|W - W_{\text{degbad}}| \Rightarrow \begin{bmatrix} 0 & 0 & 0 & 0.83 \\ 0 & 0 & 0.6 & 0 \\ 0.28 & 0 & 0 & 1.03 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Mean Absolute error for $S_{\text{Bad}} = \frac{\text{Sum of all elements of } |W - W_{\text{degbad}}|}{16}$

$$= \frac{2.74}{16}$$

$$= 0.17125$$

Finally, when S is too small like $0.01 S_{\text{bad}}$, large weights in the matrix are exceeding the INT8 range and we are clamping to $[-128, 127]$ range which is causing saturation and much larger dequantization errors.